WASP-43b

R-0581

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_230322` · created 2026-07-24T17:58:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460025.7511 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.059
Transit depth (Rp/Rs)^2	3.9 +/- 2.31 [%]
Orbital Inclination (inc)	87.1 +/- 2.87
Airmass coefficient 1 (a1)	0.47 +/- 0.16
Airmass coefficient 2 (a2)	0.52 +/- 1.26
Scatter in the residuals of the lightcurve fit is	34.35 %
Best Comparison Star	#1 - [76, 333]
Optimal Aperture	1.83
Optimal Annulus	9.93
Transit Duration (day)	0.0613 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.059 vs 0.1594 (deviation 0.65σ)
Duration fit vs published	0.0613 d vs 0.0512 d (deviation 2.05σ)
Mid-time O–C	-3.2 min
Depth SNR	3.4
Residual scatter	34.35 %

Light curve

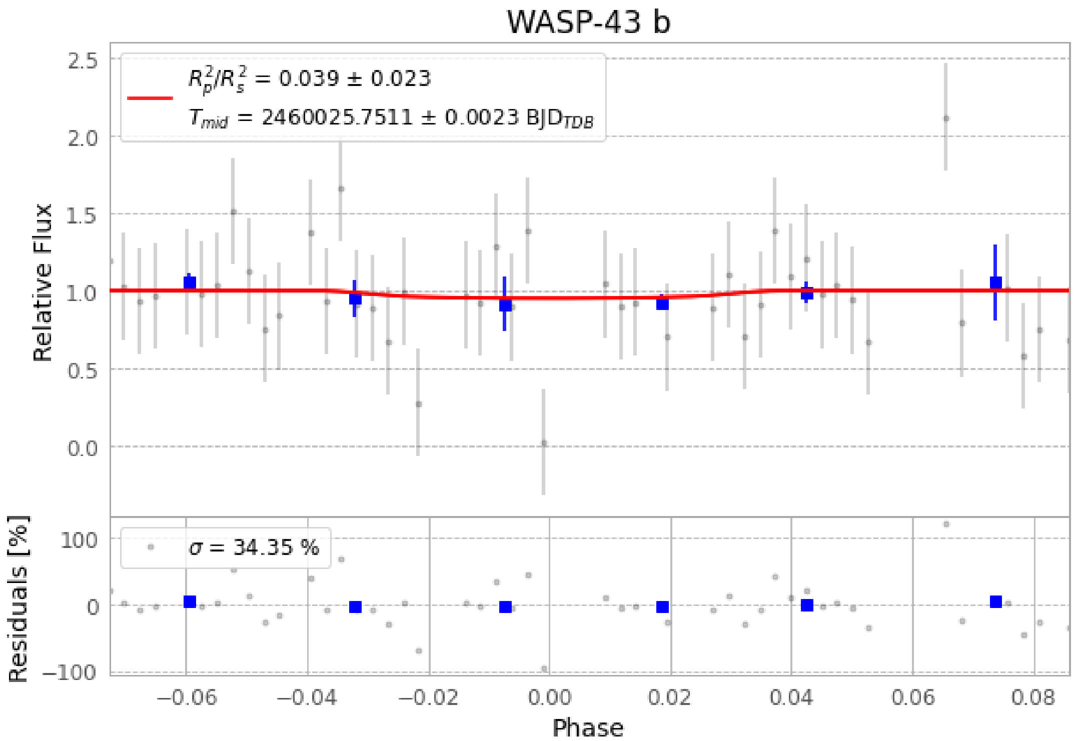

Figure 1 — FinalLightCurve_WASP-43 b_2023-03-21.png (SHA-256 a0248ebc513d81fd…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [455, 162], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e77d74d0279bf5de…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

61 FITS frames (40 MB), `WASP-43230322042717.FITS` → `WASP-43230322073315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230322.log (e6a1f1904445…), FinalLightCurve_WASP-43 b_2023-03-21.png (a0248ebc513d…), FinalLightCurve_WASP-43 b_2023-03-21.csv (e9803fedc4d1…)

Compute resources

Wall time 210.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)